Computer Networking - Quick Interview Notes

OSI Model, TCP/IP, Protocols

1. OSI 7 Layers (top to bottom)

- 7 Application, 6 Presentation, 5 Session, 4 Transport,
- 3 Network, 2 Data Link, 1 Physical.
- Mnemonic: All People Seem To Need Data Processing.

2. TCP/IP Model (4 layers)

- Application, Transport, Internet, Network Access.
- HTTP, FTP, SMTP, DNS live at Application layer.

3. TCP vs UDP

- TCP: connection-oriented, reliable, ordered, flow control.
- UDP: connectionless, fast, no guarantees (video, DNS).
- 3-way handshake: SYN, SYN-ACK, ACK.

4. Important Protocols

- HTTP/HTTPS: web (HTTPS = TLS encryption).
- DNS: resolves domain names to IP addresses (port 53).
- DHCP: auto-assigns IP addresses (port 67/68).
- SMTP: sending mail (25/587). IMAP/POP3: receiving mail.
- FTP/SFTP: file transfer. SSH: secure remote shell (22).

5. IP Addressing

- IPv4: 32-bit, e.g. 192.168.1.1 (private range).
- IPv6: 128-bit, solves address exhaustion.
- Subnet mask divides network and host portions.
- NAT: maps private IPs to one public IP.

6. HTTP Basics

- Status codes: 200 OK, 301/302 redirect, 401/403 auth,
- 404 Not Found, 500 Internal Server Error.
- GET vs POST: idempotent retrieval vs state-changing submit.
- Cookies vs Sessions: client-side storage vs server-side state.

7. Common Interview Questions

- Difference between Hub, Switch, Router.
- What happens when you type a URL? (DNS -> TCP -> HTTP -> render).
- Load balancer: distributes traffic (round robin, least connections).